Muhammad Talha Atif

Backend Engineer

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SUMMARY

Backend Engineer with 1+ year building production-grade distributed systems, secure REST APIs, and event-driven microservices. Delivered privacy-sensitive IAM infrastructure across 8+ services using Kafka, gRPC, Docker, and AWS, with hands-on Python inference services and OpenSearch observability at scale. Currently deepening Go expertise to complement a strong foundation in distributed backend engineering, cloud-native deployments, and security-first API design.

WORK EXPERIENCE

Backend Engineer *Dec '25 – Present*
MicroTechX Remote

Designed and maintained production backend services for an enterprise IAM platform spanning 8+ distributed microservices with event-driven workflows and secure API communication.

- Architected Kafka event-driven microservices, configuring dead-letter queues across 8+ distributed services.
- Designed batch-query bulk assignment API supporting 5,000+ users per operation with flat latency.
- Migrated OpenSearch index, reindexing 5M+ documents via alias switching with zero downtime.

Backend Engineer *Sep '25 – Jan '26*
Digital Upgraders LLC Remote

Built and maintained Node.js backend APIs and cloud infrastructure for a production social platform with secure authentication, real-time features, and containerised deployments.

- Optimized database indexing on feed API, reducing response time from 3 seconds to 700ms.
- Replaced non-transactional queries with database transactions, eliminating notification duplication entirely.
- Led Docker-based zero-downtime production migration to Fly.io across a live multi-service platform.

PROJECTS

Moodly | AI Experience Booking Platform | Demo

- Built Python inference service classifying emotions across 7 classes from voice and facial input.
- Designed RabbitMQ async pipeline, decoupling inference from booking service under concurrent load.
- Implemented typed error schemas, auth validation, and file cleanup across 3 production API endpoints.

Memingo | Language Learning Platform | Demo

- Resolved leaderboard bottleneck using Redis precomputed cache, dropping response from 30s to 90ms.
- Built object detection module returning vocabulary and context as a single structured API response.
- Integrated AI chatbot with session-aware context, delivering adaptive recommendations across learning sessions.

SKILLS

Backend Python, FastAPI, Java, Spring Boot, Node.js, Express.js, NestJS, REST APIs, gRPC, Go (learning)
Distributed Kafka, RabbitMQ, event-driven architecture, microservices, WebSockets, gRPC
Databases PostgreSQL, MySQL, MongoDB, Redis, SQLite, Supabase, Firebase, OpenSearch, Hibernate/JPA
Security JWT, OAuth 2.0, RBAC, rate limiting, Spring Security, input validation, secure API design
Cloud AWS (S3, RDS, EC2), Docker, Kubernetes (familiar), Fly.io, Dokploy
Observability OpenSearch, structured logging, audit trails, index alias switching, aggregation queries
DevOps GitHub Actions, CI/CD pipelines, Docker Compose, Maven, Git

Tools Postman, Swagger, Stripe, VS Code, IntelliJ IDEA

AWARDS

Runner-up, App Development Hackathon (2025) *May '25*
Organized by Punjab University

Ranked, ICPC Asia Topi Online Programming Contest (2024) *Nov '24*
Organized by International Collegiate Programming Contests

EDUCATION

Bachelor's degree in Computer Science, University of Management and Technology – UMT *Oct '21 – Aug '25*